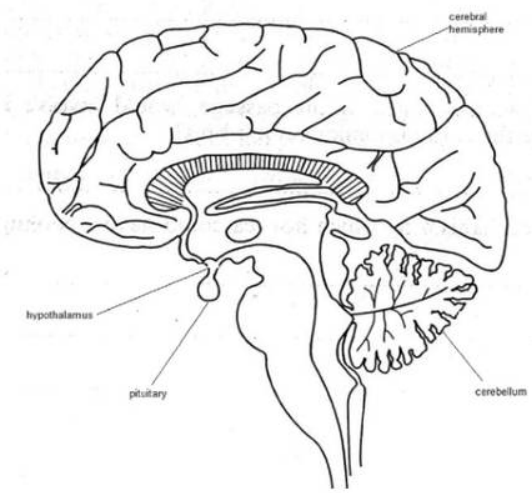
## Type MS · Identify brain regions and match them to their functions

### Answer 1 of 9 · Question 1 part b · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 1 | (a) |  correct labels ;;;; | | 4 |
| | (b) | <table><tr><th>Function</th><th>Structure</th></tr><tr><td>control of muscle tone and posture</td><td>cerebellum</td></tr><tr><td>intelligence and thought</td><td>cerebral hemisphere</td></tr><tr><td>monitoring of body temperature</td><td>hypothalamus</td></tr><tr><td>release of ADH</td><td>Pituitary</td></tr></table> | | 4 |

Total 8 marks

## Type MS · Identify brain regions and match them to their functions

### Answer 1 of 9 · Question 1 part b · 4 marks

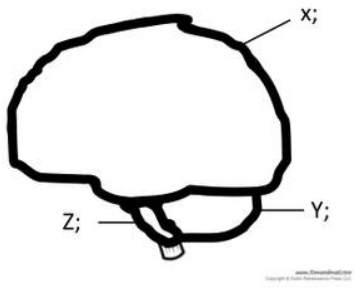
2021 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify brain regions and match them to their functions

Answer 2 of 9 · Question 7 part a · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) |  | Letters can be anywhere within outline shown.<br>Allow labels in place of than letters. | 3 |
| (b) (i) | Motor; | | 1 |
| (ii) | Any one from the following:<br><br>(sensory neurone):<br>longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord<br><br>(motor neurone)<br>shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord | Answer must be comparative and state which type of neurone is being referred to | 1 |
| (iii) | • chemicals/neurotransmitters;<br>• diffuse;<br>• across a synapse;<br>• bind to receptors (on post-synaptic membrane); | | 3 |
| (c) | • stem cells are undifferentiated/unspecialised/can differentiate;<br>• could be made to form nerve cells;<br>• to replace damaged cells in brain; | | 1<br>1<br>1 |

Total question 7 = 11 marks

## Type MS · Identify brain regions and match them to their functions

### Answer 2 of 9 · Question 7 part a · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify brain regions and match them to their functions

Answer 3 of 9 · Question 6 part b · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| **6(a)** | short term is recent event/long term goes back many years;<br>short term very little processing, long term has more processing; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(b)** | cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum; | | **(1)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(c)** | • language/speaking/talking problems / repetitive actions/confusion;<br>• loss of voluntary muscle function;<br>• loss of intellectual capacity/function;<br>• deterioration in social behaviour; | | **(2 max)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(d)** | • blocks synapse/cleft/build-up of plaques;<br>• prevents diffusion of acetylcholine / transmitter/neurones communicating;<br>• blocks post-synaptic receptors/prevents generation of new impulse;<br>• death of neurones; | | **(3 max)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **6(e)** | • inhibits/prevents breakdown of acetylcholine / transmitter;<br>• increases chance of synaptic transmission / neurones communicating/generating new impulse;<br>• reduces death of neurones; | | **(2)** |
| Question Number | Answer | Additional guidance | Mark |
| **6(f)** | formation of β-amyloid is caused by incorrect breakdown of APP;<br>build-up/formation of β-amyloid (in neurones); | | **(2)** |

**Total 12 Marks**

Type MS · Identify brain regions and match them to their functions

Answer 3 of 9 · Question 6 part b · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Identify brain regions and match them to their functions

Answer 4 of 9 · Question 7 part a.i · 3 marks

2020 JAN · Paper 02 · find part a.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 7 (a) (i) | <table><tr><th>Part</th><th>Function</th><th>Tick</th></tr><tr><td>medulla</td><td>controls heart rate and breathing</td><td>✓</td></tr><tr><td>pituitary gland</td><td>involved in language</td><td></td></tr><tr><td>hypothalamus</td><td>responsible for osmoregulation</td><td>✓</td></tr><tr><td>cerebellum</td><td>controls voluntary movements</td><td>✓</td></tr></table> | Maximum of two marks if all four boxes are ticked. | 3 |
| (ii) | • transmit impulses to effector organs/muscles/glands/controls movement/voluntary actions(1) | | 1 |
| (b) | • lack of dopamine/no dopamine produced / released/release of dopamine inconsistent(1)<br>• less nerve impulses generated/nerve impulses (to muscles/along motor neurones) inconsistent (1)<br>• tremor/rigidity/slowness of movement/imbalance occur(1) | Allow neurotransmitter for dopamine | 3 |

**Total for Question 7 = 7 marks**

Type MS · Identify brain regions and match them to their functions

Answer 5 of 9 · Question 7 part a.ii · 1 mark

2020 JAN · Paper 02 · find part a.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 7 (a) (i) | <table><tr><th>Part</th><th>Function</th><th>Tick</th></tr><tr><td>medulla</td><td>controls heart rate and breathing</td><td>✓</td></tr><tr><td>pituitary gland</td><td>involved in language</td><td></td></tr><tr><td>hypothalamus</td><td>responsible for osmoregulation</td><td>✓</td></tr><tr><td>cerebellum</td><td>controls voluntary movements</td><td>✓</td></tr></table> | Maximum of two marks if all four boxes are ticked. | 3 |
| (ii) | • transmit impulses to effector organs/muscles/glands/controls movement/voluntary actions(1) | | 1 |
| (b) | • lack of dopamine/no dopamine produced / released/release of dopamine inconsistent(1)<br>• less nerve impulses generated/nerve impulses (to muscles/along motor neurones) inconsistent (1)<br>• tremor/rigidity/slowness of movement/imbalance occur(1) | Allow neurotransmitter for dopamine | 3 |

**Total for Question 7 = 7 marks**

## Type MS · Identify brain regions and match them to their functions

### Answer 6 of 9 · Question 6 part a.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | A; (brain) | | 1 |
| | | (ii) | B; (endocrine) | | 1 |
| | (b) | (i) | • 75 (from weight of man) (1)<br>• 320 × 2 (extra from drinking beer) (1)<br>• 715cm$^3$ (1) | | 3 |
| | | (ii) | • 715 - 75 (1)<br>• (640 ÷ 75) × 100 (1)<br>• = 853% (1) | | 3 |
| | (c) | | any two from<br>• increases permeability of collecting duct (1)<br>• more water reabsorbed (1)<br>• into blood (1) | | 2 |
| | (d) | | any two from<br>• red blood cells loses water (1)<br>• become distorted (1)<br>• less oxygen transported (1)<br>• volume of plasma decreases (1)<br>• blood becomes thicker/more difficult to circulate(1)<br>• increase in blood pressure (1) | Ignore less water in blood<br>Allow water potential decreases | 2 |
| | (e) | | any three from<br>• waste products continue to be produced (1)<br>• toxic (1)<br>• need to be removed in urine (1)<br>• less reabsorption of water (1)<br>• less ADH produced (1) | Allow named waste product(s) | 3 |

Total 15 marks

## Type MS · Identify brain regions and match them to their functions

### Answer 6 of 9 · Question 6 part a.i · 1 mark

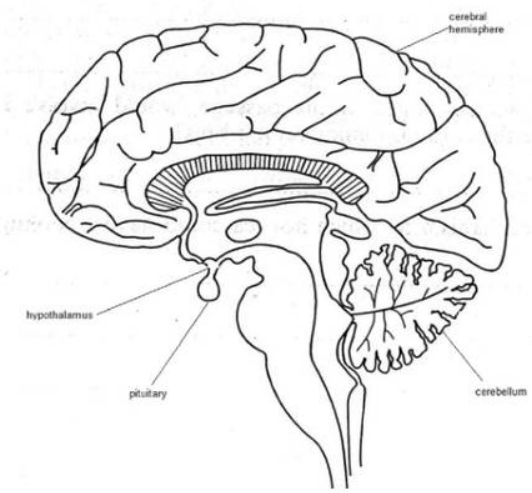
2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | Allow reverse for whole | |

## Type MS · Identify brain regions and match them to their functions

### Answer 7 of 9 · Question 1 part a · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 1 | (a) |  correct labels ;;;; | | 4 |
| | (b) | <table><tr><th>Function</th><th>Structure</th></tr><tr><td>control of muscle tone and posture</td><td>cerebellum</td></tr><tr><td>intelligence and thought</td><td>cerebral hemisphere</td></tr><tr><td>monitoring of body temperature</td><td>hypothalamus</td></tr><tr><td>release of ADH</td><td>Pituitary</td></tr></table> | | 4 |

Total 8 marks

## Type MS · Identify brain regions and match them to their functions

### Answer 7 of 9 · Question 1 part a · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify brain regions and match them to their functions

### Answer 8 of 9 · Question 3 part a · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

| number | | | |
|---|---|---|---|
| 3 (a) | C hypothalamus;<br><br>A does not control sweating<br>B does not control sweating<br>D does not control sweating | | 1 |
| (b) | B motor;<br><br>A transmits impulse from sensory to motor<br>C alternative name for A with same function<br>D transmits impulse to brain | | 1 |
| (c) | • (sweat) evaporates from (the surface of) the skin;<br>• using heat from body/blood;<br>• cools body/reduces body temperature; | | 3 |
| (d) | • runs in families;<br>• must be faulty/mutated gene/chromosome/allele;<br>• which transfers/passes on information to offspring; | | 3 |
| (e) | obesity / high sugar intake / high BMI; | | 1 |
| (f) | • exercise/activity of individual varies;<br>• greater rate of aerobic respiration;<br>• heat given off as by-product/increased body temperature;<br>• external temperature varies;<br>• hotter more sweating/cooler less sweating;<br>• mass of individual varies;<br>• volume of water in the body/fluid intake varies; | Allow reverse argument for mps 2 and 3 | 5<br><br>Total 14 |

## Type MS · Identify brain regions and match them to their functions

### Answer 8 of 9 · Question 3 part a · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify brain regions and match them to their functions

Answer 9 of 9 · Question 3 part d · 1 mark

2024 JUN · Paper 02 · find part d in this mark scheme extract

| Question number | | | |
|---|---|---|---|
| 3 (a) | Smoking cigarettes;<br>Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ;<br><br>OR<br><br>lack of exercise;<br>obesity/heart disease/high blood pressure/ type ii diabetes/eq; | Allow drugs or named drug and their effect on health for 2 marks. | (2) |
| (b) | Any two from:<br>• production of bile reduced/affected;<br>• less fat emulsified/fat not emulsified;<br>• lipase less active/lipase unable to digest as much (fat)/slower lipase activity; | Allow bile not produced<br><br>Allow description of emulsification | (2) |
| (c) | blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina; | | (1) |
| (d) | cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata); | Allow any part of the brain that is affected by alcohol | (1) |
| (e) | Any three from:<br>• reduces release of neurotransmitters;<br>• reduced diffusion across synapse;<br>• less neurotransmitters bind to receptors on post-synaptic membrane;<br>• slows/ fewer transmission of impulses;<br>• slower reaction /reflexes; | | (3) |
| (f) | Any two from<br>• increased heart rate;<br>• increased breathing rate;<br>• pupil dilation;<br>• increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose;<br>• increase in blood flow to brain/muscles;<br>• increase in (aerobic) respiration (in muscles); | | (2) |

**Total for Question 3 = 11 marks**